



Cambrian College

School of Justice, Community Services and General Studies

Course Outline

Cambrian College would like to acknowledge that we are situated on the traditional lands of the Robinson Huron Treaty, and acknowledge our host, Atikameksheng Anishnawbek, as well as our ancestors of this land, and the Anishnaabe people.

Course Title	GIS Mapping				
Course Code:	GIS1025	Credit Value:	2	Credit Hours:	28
Programs:	BAPG Business Analytics CAGC Crime Analytics CAPT Crime Analytics - Pt HAGC Health Analytics HAPT Health Analytics - Pt				
Equivalencies		Prerequisites		Corequisites	

This course may be delivered in a variety of different formats: 100% in-class, 100% online (or a blend of both), videoconferencing, distributed learning or off-campus. Please confirm with your faculty member which format will be used for your section of this course.

General Education Course: ☐

Degree Breadth Course: ☐

Eligible for PLAR: ☒

COURSE DESCRIPTION

In this course, students will be introduced to Geographic Information System (GIS) and its use as mapping, cartography, and analysis applicable to crime analyst. Students will learn geoprocessing, geocoding and GIS topology. Students will explore the capabilities of ESRI's ArcGIS software (ArcPro, ArcGlobe, and ArcScene) and ArcGIS Online. Analysis will be performed using the Crime Analyst and Spatial Analyst extensions.

Date: May 25, 2021

Prepared By: Sidney Shapiro

Approved by: _____

BRADIE GRANGER
Chair, School Of Justice, Community
Services And General Studies

Effective: Fall 2021, Winter 2022, Spring 2022

RELATIONSHIP TO PROGRAM VOCATIONAL LEARNING OUTCOMES

PROGRAM LEVEL	
This course contributes to your program by allowing you to demonstrate the following vocational learning outcomes:	
Program(s)	Vocational Learning Outcomes
Business Analytics	College Standards 1. Collect, manipulate and mine data sets to meet organizational need. 2. Design and apply data models that meet the needs of a specific operational/business process. 3. Design and present data visualizations to communicate information to business stakeholders. 4. Apply data analytics, business intelligence tools and research to support evidence-based decision making. 5. Identify and assess data analytics business strategies and workflows to respond to new opportunities or provide project solutions.
Crime Analytics	College Standards 1. Collect and analyze a variety of data sets to identify information related to crime analysis. 2. Design and apply data models that meet the needs of a specific operational process. 3. Design and present data visualizations to communicate information to stakeholders. 4. Develop a crime analytics methodology grounded in a conceptual framework and consistent with criminal intelligence techniques. 5. Evaluate the risk of criminal behavior based on an interpretation of data. 6. Conduct crime analysis and related research while maintaining strong ethical standards and an awareness of moral and ethical implications in a confidential workplace.
Crime Analytics - Pt	College Standards 1. Collect and analyze a variety of data sets to identify information related to crime analysis. 2. Design and apply data models that meet the needs of a specific operational process. 3. Design and present data visualizations to communicate information to stakeholders. 4. Develop a crime analytics methodology grounded in a conceptual framework and consistent with criminal intelligence techniques. 5. Evaluate the risk of criminal behavior based on an interpretation of data. 6. Conduct crime analysis and related research while maintaining strong ethical standards and an awareness of moral and ethical implications in a confidential workplace.
Health Analytics	College Standards

Program(s)	Vocational Learning Outcomes
	1. Collect and analyze a variety of data sets to identify information related to organizational health needs. 2. Mine and manipulate data sets, correlate information and produce outputs using advanced software functions. 3. Design visualizations and communicate results of data analysis to decision makers, stakeholders and the public. 4. Apply health analytics and business intelligence tools to support evidence-informed practice. 5. Synthesize relevant local, regional, provincial, national and global health and health information management issues, trends, technologies and standards to support health information systems and processes. 6. Ensure the completeness, accuracy, consistency, timeliness and integrity of health information throughout the management of its lifecycle.
Health Analytics - Pt	College Standards 1. Collect and analyze a variety of data sets to identify information related to organizational health needs. 2. Mine and manipulate data sets, correlate information and produce outputs using advanced software functions. 3. Design visualizations and communicate results of data analysis to decision makers, stakeholders and the public. 4. Apply health analytics and business intelligence tools to support evidence-informed practice. 5. Synthesize relevant local, regional, provincial, national and global health and health information management issues, trends, technologies and standards to support health information systems and processes. 6. Ensure the completeness, accuracy, consistency, timeliness and integrity of health information throughout the management of its lifecycle.

COURSE CURRICULUM

Topics/Concepts Covered in This Course

- Understanding Geographic Information System (GIS)
- Discovering GIS
- Creating spatial databases (geodatabases and shapefiles) and join spatial and non-spatial tables
- Current uses of ESRI ArcGIS Pro along with Crime Analyst and Spatial Analyst extensions
- Learning geoprocessing, geocoding, and GIS topology
- Understanding coordinates systems, map projections, and cartography
- Generating Modelbuilder and ArcPy modules
- Current uses of ESRI ArcGIS Online
- Constructing story maps, web maps, web applications, operations dashboards, and mobile applications
- Introduction to ESRI ArcGlobe, ArcScene, GPS data, and open source data

COURSE LEVEL: Learning Outcomes and Objectives	
To earn credit for this course, you must reliably demonstrate your ability to:	
Learning Outcome	Objectives
1. Use the ESRI ArcGIS Pro software	1.1 Recognize the capability of GIS to solve real-world problems 1.2 Execute queries, analyze, and geoprocess GIS data 1.3 Recognize GIS data (vector and raster) and understand how to create them 1.4 Demonstrate topology in GIS 1.5 Perform geoprocessing and geocoding
2. Generate and interpret maps	2.1 Apply graphic design in making a map 2.2 Identify the map elements in creating a map
3. Create and manage databases of geo-referenced information	3.1 Discuss and manipulate the data contents of a database for use of GIS 3.2 Practice to join and relate spatial and non-spatial tables
4. Apply different coordinate systems and map projections	4.1 Practice in converting projections in ESRI ArcGIS Pro 4.2 Practice how to convert lat/long to UTM 4.3 Practice the conversion tools in ArcTools
5. Apply Crime Analyst and Spatial Analyst ArcGIS extensions	5.1 Apply a series of key analysis functions 5.2 Demonstrate the different methods used in performing certain analysis
6. Generate ModelBuilder and ArcPy modules	6.1 Create a ModelBuilder module 6.2 Compute ArcPy syntax, commands, built-in functionalities, and functions
7. Use ESRI ArcGlobe, ArcScene, GPS data, and open source data	7.1 Manage and apply data existing on open source data websites 7.2 Transfer and use data from a GPS source into GIS 7.3 Visualize 3D data in ArcGlobe and ArcScene
8. Apply the ESRI ArcGIS Online applications	8.1 Design story maps, web maps, web applications, and operations dashboards 8.2 Produce mobile applications

Essential Employability Skills

Communication

- communicate clearly in written, spoken, and visual form that fulfills purpose/needs of audience.
- respond to written, spoken, or visual messages in a manner that ensures effective communication.

Numeracy

- execute mathematical operations accurately.

Critical Thinking and Problem Solving

- apply a systematic approach to solve problems.
- use a variety of thinking skills to anticipate and solve problems.

Information Management

- locate, select, organize, and document information using appropriate technology and info systems.
- analyze, evaluate, and apply relevant information from a variety of sources.

Interpersonal

- show respect for the diverse opinions, values, belief systems, and contributions of others.
- interact with others in groups in ways that contribute to effective working relationships.

Personal

- manage the use of time and other resources to complete projects.
- take responsibility for one's own actions, decisions, and consequences.

Delivery Method

- Classroom: Course is delivered through scheduled synchronous teaching that may be face-to-face and/or virtual.
- Online: Course is fully delivered through asynchronous teaching.
- Hybrid: Course combines scheduled synchronous and unscheduled asynchronous teaching.
- HyFlex: Course includes both synchronous and asynchronous learning and the student can move between both components seamlessly.

Learning Activities

- Lectures
- Labs
- Research
- Self-Directed Learning
- In-Class Exercises
- eLearning Components

Resources Required

Additional Supplies

N/A

All materials will be supplied by ESRI Canada (The Education and Research group) and through PowerPoint Presentations

Evaluation Plan

Grading Scheme

A	80% - 100%
B	70% - 79%
C	60% - 69%
D	50% - 59%
F	0% - 49%

Evaluation Method	Value (%)
Assignments	30%
There are multiple assignments in this course, please contact your instructor for more information	
Test/Exam	70%
This course has multiple assessments. Please see your instructor for details.	

ADDITIONAL INFORMATION

College

Modifications to the Course Evaluation Plan

Under exceptional circumstances, Cambrian reserves the right to alter the Course Evaluation Plan. Upon approval from the Dean, faculty members will notify students of any modification to the Course Evaluation Plan and post to the course Learning Management System site (Moodle).

Academic Policies

Students must adhere to all academic policies. These are available on myCambrian as well as the College website at <https://cambriancollege.ca/about/official-documents-and-policies/academic-policies/>.

Accessibility Services

Cambrian is committed to creating an inclusive learning environment where we support our students' success. Students who are registered with the Glen Crombie Centre can access accessibility and/or counselling services by:

1. Registering and scheduling an appointment through Clockwork in the Student Portal in myCambrian.
2. Contacting 705-566-8101 ext. 7420 or 7311, or
3. E-mailing accessibilityservices@cambriancollege.ca or Counselling@cambriancollege.ca

Attendance

Regular attendance is strongly encouraged as it contributes to student success. Some courses have specific lab/participation requirements. Students are expected to be aware of, and adhere, to these requirements.

Audio/Visual Capture

Sounds and images from this class, and contributions made by a participant, virtually or in-person, are recorded under the authority of the Ontario Colleges of Applied Arts and Technology Act, 2002. The main purpose for these recordings is to allow students enrolled in this course, whether they attend the particular class in person, virtually, or at all, to review class content and activity. The use of class recordings is for personal use only and shall not be shared or transferred.

These recordings may also be reviewed by faculty members to prepare future classes, to evaluate students, to collaborate in program/course development/review, or to provide feedback to students or other faculty. Any questions about this collection may be addressed to your respective Dean.

Copyright

Students may not submit work or partake in activities that infringe on the Canadian Copyright Act, or the related Cambrian College Fair Dealing Guidelines and Copying Guidelines. The guidelines may be viewed at: <http://cambriancollege.libguides.com/copyrightinfo> or by contacting the Library for questions related to copyright: library@cambriancollege.ca

Wabnode Centre for Indigenous Services

Students are encouraged to come to Wabnode to meet our Elders and to access personal counselling and connect with academic tutoring. Our team is also able to provide students with information related to funding and scholarship and bursary opportunities for Indigenous students. We encourage students to come to Wabnode and find out about opportunities to share in cultural, sports and community activities. Students can also contact Wabnode at wabnode@cambriancollege.ca

Out-of-Class Assistance

The faculty will inform students of their Out-of-Class availability through the Learning Management System site (Moodle).

Prior Learning Assessment and Recognition (PLAR)

Students wishing to have work or life experience considered through Prior Learning Assessment and Recognition should contact pathways@cambriancollege.ca

Testing Practices at Cambrian

Many courses include major tests and/or final exams. The practice at Cambrian requires that these types of test situations involve proctoring to ensure academic integrity. Online tests/exams may employ a proctoring service to enable you to take your exam from a location of your choosing within a period specified by your instructor. When you are taking an online test/exam, the proctoring service may capture your video, screen, audio and web surfing particulars to protect academic integrity. Cambrian College collects, uses, discloses and retains personal information in compliance with the Freedom of Information and Protection of Privacy Act (FIPPA). Your personal information is being collected under the authority of the Ontario Colleges of Applied Arts and Technology Act S.O. 2002, c.8, Sched. F. This information will be used for the purpose of administering a test/exam through an online proctoring service acting as an authorized agent of the College. Please refer to Cambrian's Confidentiality of Student Records Policy for more details. If you have any questions regarding the collection of your personal information, please contact Vice President Academic, Cambrian College, 1400 Barrydowne Rd., Sudbury ON P3A 3V8, 1-705-566-8101.

Tests and Evaluations

Students are required to write tests and/or complete evaluations as scheduled. Exceptions may be made in the event of an emergency or a sanctioned event (e.g., varsity sports, field trip, religious observances, etc.). In non-emergency situations, students are expected to contact the faculty member in the event that they can not be present for a test/evaluation to explain the reason for their absence. At the discretion of the faculty member, an alternate time for the test/evaluation may be allowed.

Transfer Credit

Students wishing to have courses from other programs or institutions assessed for equivalency and/or transfer credit should contact transfercredits@cambriancollege.ca.

Equity, Diversity and Inclusivity

Cambrian is committed to building and preserving an equitable, diverse, and inclusive learning community where students, faculty, and staff may achieve their full potential in an environment characterized by equality of respect and opportunity. All students and employees have the right to live and work in an environment that is free from discrimination and harassment. Therefore, Cambrian College will not tolerate any form of discrimination or harassment in its employment, education, accommodation, or business dealings. For more information, please visit: <https://cambriancollege.ca/about/official-documents-and-policies/equity-human-rights-and-accessibility/>